

Digital Platform for Small and Home-Based Entrepreneurs

HACKATHON PRODUCT BLUEPRINT, SYSTEM ARCHITECTURE & IMPLEMENTATION GUIDE

Engineered for Rapid Hackathon Prototype Delivery | Date: October 2026

Executive Overview: This technical blueprint translates the challenge of empowering home-based micro-entrepreneurs into a functional, award-winning hackathon prototype. While traditional platforms operate as closed retail marketplaces, this solution serves as an intelligent business-enablement and opportunity-matching engine. It natively unifies tangible goods and skilled service providers, connects them to retail buyers, B2B institutions, external marketplaces via open protocols (ONDC), peer consortiums, working capital financing (PO escrow/OCEN), and proactive government schemes.

1. De-Bloating & Refinement: What Was Removed

To make the product viable for a focused hackathon execution, extensive theoretical and corporate bloat from earlier exploratory reports was removed:

- **Macroeconomic & Global Statistics:** Pruned multi-trillion-dollar macro statistics (e.g., general World Bank and OECD quotes) that distract from the practical user problem.
 - **Speculative Multi-Year Corporate Roadmaps:** Removed theoretical 5-phase rollouts spanning multiple years; replaced with an actionable 48-hour build timeline.
 - **Passive Directory / Classifieds Logic:** Eliminated static browse-and-contact listings. The platform is refocused on active, programmatic matching and automated quotations.
 - **Deferred Logistics & Finance Clauses:** Overturned previous recommendations to delay finance and shipping. Hackathon judges expect working mechanics; the MVP includes immediate working prototypes for milestone escrow, PO-based credit checks, and hyperlocal dispatch.
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2. Updated Core Functional Modules

The platform addresses every requirement of the problem statement through seven interconnected modules:

A. Dual Engine: Goods & Skilled Services

Micro-entrepreneurship spans two distinct operational models. The platform handles both seamlessly:

- **Physical Goods (Food, Tailoring, Handicrafts, Handmade Items):** Accommodates batch inventory, packaging constraints, perishable shelf life (e.g., home bakery/tiffin), and Minimum Order Quantities (MOQ).

- **Skilled Services (Beauty, Tutoring, Repair Services):** Manages calendar appointment slots, service delivery modes (at home, client site, or remote/online for tutoring), travel radius (km), diagnostic/visitation fees, and verifiable skill certifications.

B. Resource & Capacity Profiling

Beyond static skill tags, matching requires assessing physical resources:

- **Equipment & Tools:** Captures machinery availability (e.g., industrial lockstitch sewing machine, 4-tier commercial oven, multimeter testing gear).
- **Workspace & Regulatory Readiness:** Logs commercial kitchen status (FSSAI), dust-free assembly areas, and storage capacity.
- **Working Capital Liquidity:** Tracks upfront capital constraints so orders can be paired with advance funding.

C. Peer Partnerships: Consortium Bidding & Group Buying

Individual home businesses frequently turn down bulk opportunities due to limited solo capacity. The platform resolves this through collaborative mechanisms:

- **Consortium Pooling:** When an institution requests 600 school uniforms, the algorithm splits the order across three verified local tailors with compatible equipment, generating a single consolidated B2B quotation.
- **Raw Material Group-Buying:** Aggregates demand across multiple artisans or food producers to buy fabric, packaging boxes, or ingredients in bulk directly from mills and wholesalers at discount rates.

D. Multi-Channel Syndication & ONDC Connectivity

Instead of forcing sellers onto a captive island app, the platform acts as a Seller Network Participant (SNP) on **ONDC (Open Network for Digital Commerce)**. Catalog items are broadcast across national buyer apps (Paytm, Magicpin, Mystore), while social commerce integration exports catalogs directly into WhatsApp Business.

E. Working Capital Financing & Milestone Escrow

Solves the cash-flow trap where receiving a large order threatens survival due to upfront material costs:

- **Buyer Advance Escrow:** Institutional buyers deposit funds into escrow. Upon order confirmation, 30% is immediately unlocked to the entrepreneur exclusively for verified raw material procurement.
- **Digital PO Financing (OCEN Architecture):** Confirmed platform purchase orders generate a digitally verifiable invoice, allowing instant micro-lender working capital advances.

F. Proactive Government Scheme Rules Engine

Rather than linking out to external web portals, the engine computes eligibility automatically:

- **PM Vishwakarma:** Matched for traditional artisans, tailors, and repair craftsmen (unlocking ₹15,000 tool-kit incentives and 5% collateral-free credit).

- **PM SVANidhi & MUDRA (Shishu):** Flagged for working capital micro-loans up to ₹50,000 without collateral.
- **Udyam / Udyam Assist:** Generates pre-filled registration payloads to formalize informal home units in one tap.

G. Hyperlocal & Cluster Logistics

Integrates on-demand point-to-point courier APIs (Porter, Dunzo API mock) for 2–7 km local deliveries (crucial for home-cooked meals and urgent repairs) while supporting cluster-based pickup points for regional artisanal shipments.

3. System Architecture & Data Flow

The architecture emphasizes clean separation of concerns, lightweight client interfaces for low-bandwidth environments, and high-performance semantic search.

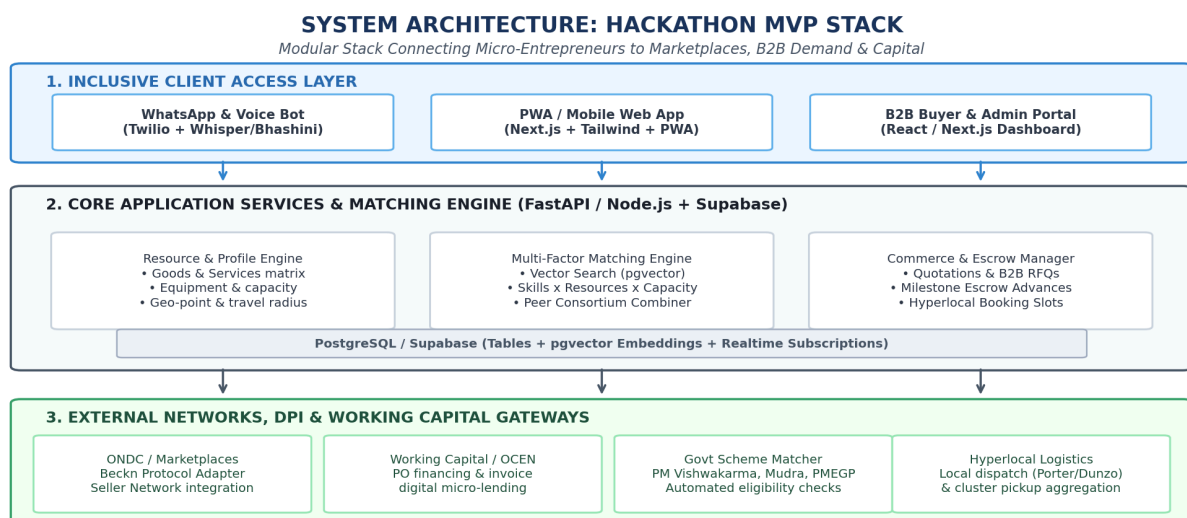


Figure 1: High-Level System Architecture and Digital Public Infrastructure Integration Stack

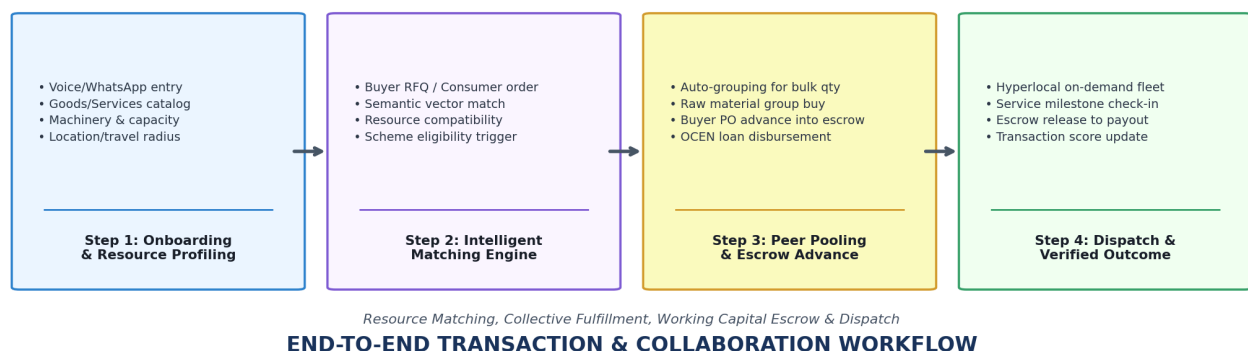


Figure 2: End-to-End User Transaction, Consortium Pooling, and Escrow Fulfillment Flow

4. Core Data Models & Matching Algorithm

Database Schema (PostgreSQL with pgvector)

Entity	Key Fields & Schema Types	Description & Purpose
Entrepreneurs	id (uuid), name, phone, type (GOODS SERVICES), geo (point), travel_radius_km, rating, verified_level	Core identity record supporting geo-spatial queries and dual-persona operations.
Resources	id, entrepreneur_id, machinery_list (text[]), workspace_type, weekly_capacity, certifications (jsonb)	Physical asset inventory used for strict feasibility verification during B2B matching.
Catalogs	id, entrepreneur_id, title, category, price, moq, delivery_type, embedding (vector(1536))	Itemized products and services with pre-computed vector embeddings for semantic search.
Orders & RFQs	id, buyer_id, title, required_qty, budget, deadline, delivery_geo, escrow_status, consortium_flag	B2C purchases and B2B requests for quotation with milestone escrow state tracking.
Consortium Allocations	id, rfq_id, entrepreneur_id, allocated_units, material_advance_paid, completion_status	Tracks order splitting across multiple home entrepreneurs collaborating on large orders.
Government Schemes	id, scheme_name, eligible_trades, max_turnover, max_loan_amount, required_docs (text[])	Eligibility criteria repository for real-time automated profile qualification.

Multi-Factor Matching Formula

Buyer queries and RFQs are matched against entrepreneur profiles using a multi-factor score:

```
MatchingScore = (0.35 × SemanticVectorSimilarity(BuyerQuery, CatalogEmbeddings))
                + (0.25 × CapacityFeasibility(RequestedQty, WeeklyAvailableCapacity))
                + (0.20 × GeoProximityScore(BuyerLocation, SellerLocation, TravelRadius))
                + (0.10 × ResourceMatch(RequiredMachinery, VerifiedEquipment))
                + (0.10 × TrustMetric(FulfilledOrders, OnTimeRate, DisputeRatio))
```

If `RequestedQty > IndividualCapacity`, the consortium allocation engine triggers automatically, clustering nearby sellers whose aggregate capacity fulfills the order.

5. 48-Hour Hackathon Build Roadmap

A prioritized sprint breakdown designed to deliver a demonstrable, working prototype within a standard hackathon duration:

Sprint Window	Development Focus	Key Deliverables & Working Components
Sprint 1 (Hours 00–12)	Foundation, Schema & Assisted Onboarding	• Initialize Supabase project, enable pgvector and PostGIS extensions. • Build responsive PWA shell (Next.js 14 + Tailwind CSS). • Prototype voice/WhatsApp onboarding using OpenAI Whisper / Bhashini audio transcription into structured JSON profiles. • Seed database with realistic artisan, baker, tailor, tutor, and beautician profiles.
Sprint 2 (Hours 12–24)	Catalog, Resources & Semantic Matching	• Build resource & capacity profiling forms (machinery, workspace, hours). • Implement buyer search interface with natural language inquiry. • Code backend matching endpoint calculating vector similarity + proximity. • Return ranked seller cards with verified capacity and equipment badges.
Sprint 3 (Hours 24–36)	Consortium Pooling, Escrow & Schemes	• Develop B2B RFQ flow: auto-grouping algorithm splitting 500 units across 3 sellers. • Implement milestone escrow simulation (Advance release for raw materials). • Create Government Scheme Matcher rule engine (evaluating profile against PM Vishwakarma, Mudra, and SVANidhi criteria). • Mock ONDC Beckn protocol search/on_search response JSON payload.
Sprint 4 (Hours 36–48)	UI Polish, Demo Scenarios & Pitch Prep	• Polish mobile and dashboard UI; ensure crisp typography and responsive layouts. • Integrate interactive map view (Leaflet / Mapbox) showing hyperlocal service radius. • Rehearse end-to-end demo script with pre-staged live test accounts. • Finalize slide deck and live interactive demonstration environment.

6. Recommended Hackathon Tech Stack

- **Frontend Web & Mobile:** Next.js 14 (App Router), React, Tailwind CSS, Lucide Icons, Shadcn UI. Fast to scaffold, responsive, and easily exportable as a Progressive Web App (PWA).
- **Backend & Database:** Supabase (Managed PostgreSQL, Authentication, Realtime subscriptions, and [pgvector](#) for semantic catalog matching).
- **Voice & WhatsApp Integration:** Twilio API / Meta Cloud API webhook for WhatsApp; OpenAI Whisper API for transcribing low-literacy voice notes into structured listing data.

- **Mapping & Hyperlocal Geolocation:** Leaflet.js / OpenStreetMap (free, no API keys required for hackathons) for displaying local delivery ranges and service radius.
 - **DPI / Network Protocols:** Mock Beckn Protocol JSON schemas for ONDC seller integration; simulated Account Aggregator (AA) data consent flow for OCEN micro-lending.
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7. Hackathon Demo Script & Winning Pitch Structure

Judges score hackathon projects on *problem significance*, *technical complexity*, *user empathy*, and *market viability*. Deliver the presentation in five concise minutes:

1. The Hook (Minute 0:00–0:45): The Capacity Trap

Introduce "Sunita," a skilled home tailor. She makes high-quality uniforms but cannot reach buyers outside her neighborhood. When a local school asks for 400 uniforms, she must decline: she lacks the single-handed capacity and cannot afford ₹60,000 for upfront fabric. Existing marketplaces only serve registered, bulk-inventory merchants.

2. Live Demo Step 1: 30-Second Voice Onboarding (Minute 0:45–1:45)

Show the entrepreneur onboarding screen. Send a voice recording via the web interface in Hindi/regional language: *"I make school uniforms and blouses. I have 2 sewing machines and can make 40 uniforms a week."* The system parses the voice, creates an instant digital profile, verifies machinery, and generates an optimized catalog.

3. Live Demo Step 2: The Impossible Bulk Order & Consortium Split (Minute 1:45–3:00)

Switch to the B2B School Procurement screen. The buyer requests 300 uniforms. The matching engine identifies that no single local tailor has 300-unit capacity before next week. It immediately triggers the **Consortium Engine**, bundling Sunita and two neighboring tailors into a joint fulfillment squad with unified pricing.

4. Live Demo Step 3: Working Capital Escrow & PM Vishwakarma Match (Minute 3:00–4:00)

Show the order confirmation. The buyer deposits payment into escrow; 35% is immediately disbursed to the tailors for raw material fabric purchases. Simultaneously, the platform's Scheme Engine alerts Sunita that she qualifies for a ₹15,000 modern toolkit grant under PM Vishwakarma, offering a one-click pre-filled application.

5. Business Model & Impact Close (Minute 4:00–5:00)

Demonstrate sustainability: 2.5% transaction commission on completed B2B sales, zero upfront fee for home entrepreneurs, and nationwide distribution through ONDC. Conclude with the mission: turning unorganized home talents into recognized, bankable micro-enterprises.